

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 9



**ECOLOGY
&
ENVIRONMENT-2**



5 TO 7% BIOMASS PELLETS ARE MANDATED IN THERMAL POWER PLANTS !

- As per the Revised Biomass Co-firing Policy of Ministry of Power dated October 8, 2021, all thermal power plants have been directed to use 5%-7% biomass pellets in their plants along with coal.
- Use of biomass in thermal power plants will not only enable generation of green electricity, will also cut down coal consumption, create employment opportunities, reduce CO₂ emissions, reduce air pollution and help in achieving India's global commitment towards 45% emissions reduction by 2030.

WHAT ARE THE CHALLENGES IN ACHIEVING 5-7% CO-FIRING?



- The availability and quality of biomass vary across regions in India. While some regions have an abundance of biomass, others face a shortage.
- Moreover, the quality of biomass also varies, which can affect its combustion efficiency and emissions.
- Biomass pellets are difficult to store at plant locations for extended periods of time because they quickly collect moisture from the air, making them unusable for co-firing. Only pellets with a moisture content of less than 13-14% typically can be burned alongside coal.
- Biomass co-firing requires specialized equipment, such as biomass grinders, conveyors, and storage systems. Additionally, power plants need to be retrofitted to enable biomass co-firing.

The transport and storage of biomass can be challenging, especially in areas where infrastructure is inadequate.

A BRIEF ABOUT THE FUNCTIONING OF WASTE TO ENERGY PROJECTS



- Waste-to-energy projects use non-recyclable dry waste to generate electricity.
- Of the non-biodegradable dry waste, only around 2-3% – including hard plastics, metals, and e-waste – is recyclable. The remainder consists of low-grade plastic, rags, and cloth that cannot be recycled.
- The waste is combusted to generate heat, which is converted into electricity.
- The process increases the State's power generation capacity and eases the Solid Waste Management (SWM) burden.

WHAT ARE THE MAJOR REASONS FOR THE FAILURE OF WASTE-TO-ENERGY PLANTS ACROSS THE COUNTRY?

1 LOW CALORIFIC VALUE

- The primary reason is the low calorific value of solid waste in India due to improper segregation. The calorific value of mixed Indian waste is about 1,500 kcal/kg, which is not suitable for power generation.
- The calorific value of segregated and dried non-recyclable dry waste is much higher at 2,800-3,000 kcal/kg, sufficient to generate power.
- Biodegradable waste has high moisture content and can't be used for power generation, it should be composted instead.

2 HIGH COSTS OF PRODUCTION

- The second reason is the high costs of energy production. The cost of generating power from waste is around Rs. 7-8/unit, while the cost at which the States' electricity boards buy power from coal, hydroelectric, and solar power plants is around Rs. 3-4/unit.



INDIAN COTTON IS PREDOMINANTLY TRANSGENIC

FACTS ABOUT INDIAN COTTON

- Primarily **transgenic** with hybrids.
- Two genes from the soil bacterium, **Bacillus thuringiensis** (Bt) transferred into the hybrid varieties.
- As hybrids **lose their genetic stability**, farmers cannot reuse the seeds. They have to **purchase** every year.
- Majority of the countries grow **open pollinated varieties** rather than hybrids.



WHY ARE FARMERS ILLEGALLY CULTIVATING HT Bt COTTON?

- It is to **reduce** the **deweeding costs**.
- With HT Bt Cotton, they can use **Glyphosate** to kill the weeds. Hence they can **reduce** the **labour costs**.

'PERMAFROST' IS THAWING

WHAT EXACTLY DO YOU UNDERSTAND BY 'PERMAFROST'?



- **Permafrost** is any ground that remains completely frozen for at least two years straight. We can say, the temperature is **-32°F (0°C) or colder.**
- These permanently frozen grounds are most common in regions with high mountains and higher altitudes near the **north and south poles.**
- Almost a quarter of the land area in the northern hemisphere has permafrost **underneath.** Although the ground is frozen, permafrost regions are **not always covered in snow.**

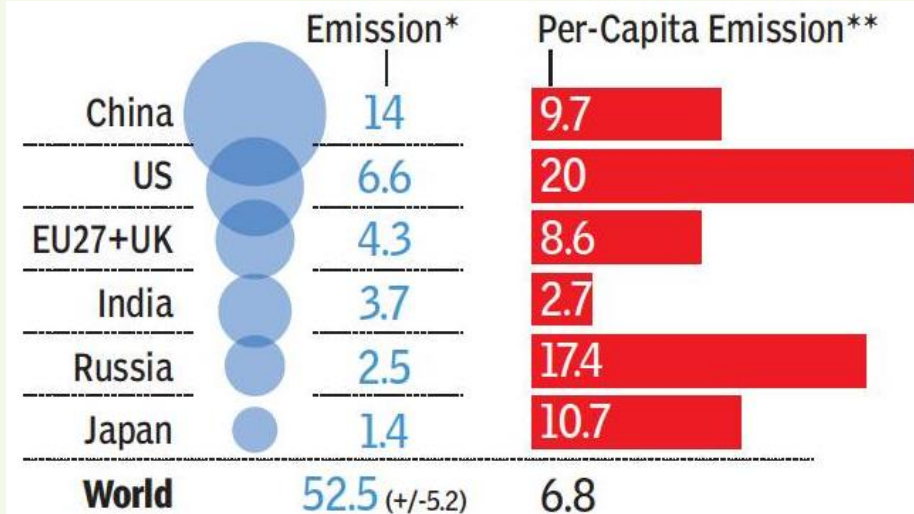
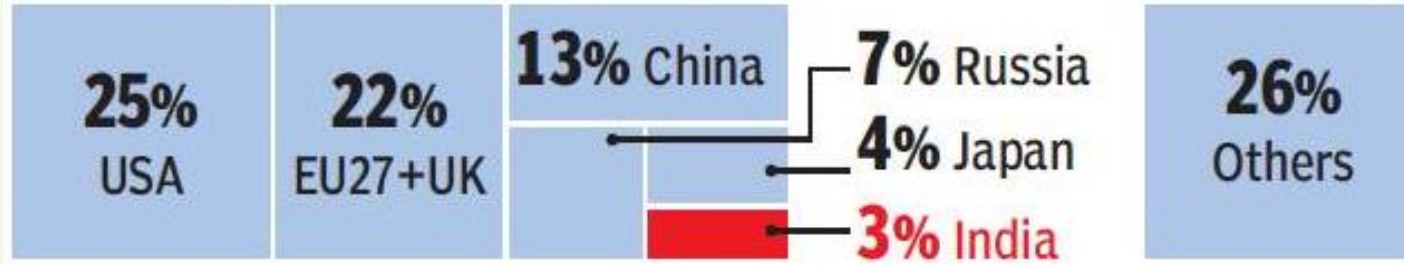
Permafrost is made up of a combination of soil, rocks and sand that are held together by ice. The soil and ice in permafrost stay frozen all year long.

WHAT HAPPENS WHEN THE CLIMATE CHANGE AFFECTS 'PERMAFROST'?



- 1) Ice inside the permafrost **melts** and leaves behind water and soil. Several villages are built on permafrost as when it is frozen, it is harder than concrete. **Thawing permafrost can destroy houses, roads and other infrastructure.**
- 2) When permafrost is **frozen**, plant material i.e. **organic carbon cannot decompose**. When permafrost thaws microbes begin decomposing this material and this releases **green house gases** like carbon dioxide and methane into the atmosphere.
- 3) When the permafrost thaws, the **newly unfrozen microbes** could make humans and animals very sick. Scientists have discovered **microbes** more than **4 lakh years old** in thawed permafrost.

HISTORICAL EMISSIONS!



In places where the maximum temperature reaches 45°C or more, the IMD declares a heatwave irrespective of the normal.



WHAT IS THE DEFINITION OF HEATWAVE?

- IMD declares a **heatwave** based on temperature.
- India Meteorological Department does not consider a heatwave unless the maximum temperature crosses **40°C in plains and 30°C in the hills.**
 - Where the normal maximum is **40°C or less**, the heatwave departure from normal is **5°C to 6°C** and the **severe** heatwave departure is **7°C or more.**
 - Where the normal maximum is more than **40°C** , the heatwave departure from normal is **4°C to 5°C** , while the severe heatwave departure is **6°C or more.**



WHAT IS WET BULB TEMPERATURE?

- The human body reacts to a combination of heat and humidity, known as the 'Wet Bulb Temperature'.
- If the temperature is 35°C , and the relative humidity is 80%, the wet bulb temperature will be 32°C , which is considered dangerous for manual labour.
- Similarly, if the temperature is 40°C , and humidity is 75%, Wet Bulb Temperature will be about 36°C .
- Few humans can tolerate a wet bulb temperature exceeding 35°C , because their bodies can no longer cool themselves.

WHAT IS INTER-GOVERNMENTAL PANEL ON CLIMATE CHANGE?

5 The AR1 formed basis for negotiation of UNFCCC in 1992. The first report was published in 1990. The AR2 was published in 1995, and it was the scientific underpinning for Kyoto Protocol in 1997. In 2007, IPCC won the Nobel Peace Prize.

ipcc
INTERGOVERNMENTAL PANEL ON
climate change



- 1 IPCC was set-up in 1988 by the World Meteorological Organization and the UN Environment Programme.
- 2 It does not itself engage in scientific research.
- 3 It asks scientists from around the World to go through all the relevant scientific literature related to climate change and drew up the logical conclusions.
- 4 The IPCC released 6 Assessment Reports.

WHAT THE SUPREME COURT SAID IN M.K. RANJITHSINH CASE?

- The Governments of Rajasthan and Gujarat shall take steps to install **bird diverters** wherever **overhead lines** in **power projects** exist. This is the first step pending consideration of the conversion of overhead cables into **underground power lines**.



FIREFLY BIRD DIVERTERS

GREAT INDIAN BUSTARD

- As per IUCN, it is **critically endangered**, hardly about **200** are **alive** in India today.
- Its **habitat** straddles across **India** and **Pakistan**, major abode is **Desert National Park**.
- The Government said **no GIB exists** in **Kutch Bustard Sanctuary** in Gujarat.
- It is facing threats due to **overhead power lines** due to its **poor frontal vision**.

Flying into power transmission lines and windmills is a major factor behind the bustard's decline.

A BRIEF ABOUT BATS

- They have a unique **sonar-based echolocation mechanism** to capture prey at night.
- Because of their ability to **navigate** and **fly in darkness**, several Batman stories came to the forefront, which are popular among children.
- They also gave rise to **baseless myths** and **unfounded fears** leading to their persecution.



WHERE DO WE FIND BATS?

- Bats are the **largest mammalian group** after rodents. They have **1,300 species** making up a quarter of all mammals.
- They occur on **all continents except Antarctica** and are particularly diverse in South Asia: there are **114 species** of **insect-eating bats** and **14 fruit bats** in South Asia.



CENTRE AND STATES DERIVE THEIR POWERS FROM NDMA



- The National Disaster Management Act (NDMA), 2005 is the key piece of legislation governing the roles of the Centre and States in responding to a natural disaster.
- The Ministry of Home Affairs is the nodal body that governs the execution of this Act.
- Several disaster management authorities draw their powers from this legislation and they define which natural calamities qualify for state backed compensation.
- The legislation is also the genesis of special funds - at the State-level and the Centre - that can be drawn upon for a disaster.

THERE ARE TWELVE NOTIFIED DISASTERS

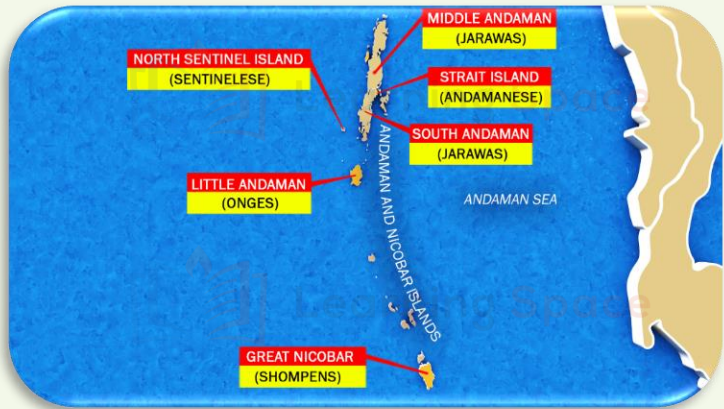


- However, in terms of the present guidelines, a State Government can use up to 10% of the annual fund allocation of the SDRF, with certain caveats, for providing immediate relief to the victims of natural disasters other than those notified 12 disasters.

- At present, 12 disasters are notified in the Guidelines on Constitution and Administration of the State Disaster Response Fund (SDRF) and National Disaster Response Fund (NDRF).
- These are:
 - ✓ Cyclone.
 - ✓ Drought.
 - ✓ Earthquake.
 - ✓ Fire.
 - ✓ Flood.
 - ✓ Tsunami.
 - ✓ Hailstorm.
 - ✓ Landslide.
 - ✓ Avalanche.
 - ✓ Cloudburst.
 - ✓ Pest attack.
 - ✓ Frost & Cold waves.

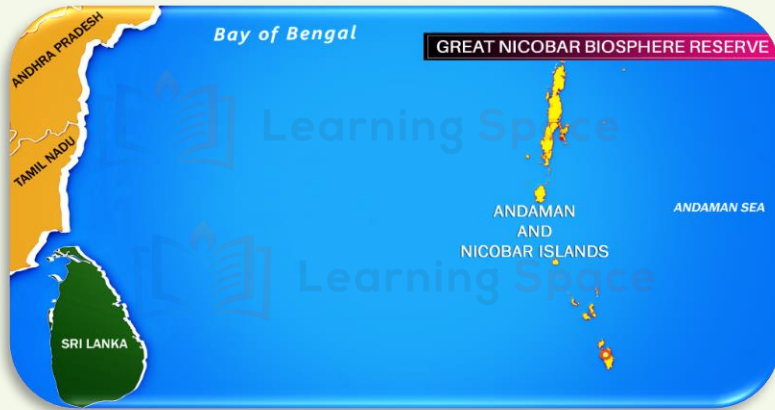
Heatwaves are yet to be included in the list of notified disasters.

A BRIEF ON GREAT NICOBAR ISLAND



- The island of Great Nicobar is the Southernmost tip of India and a part of the Andaman and Nicobar archipelago that comprises 600-odd islands.
- It is covered with lush rainforests that are sustained by around 3,500 mm of annual rainfall.
- It has an area of 910 sq. km. with mangroves and Pandan forests along its coast.
- It is home to two tribal communities - the Shompen and the Nicobarese.
- The Shompen, around 250 in total, are predominantly hunter-gatherers and are classified as a PVTG within the list of Scheduled Tribes.

MORE ASPECTS ABOUT THE GREAT NICOBAR ISLAND



- The Nicobarese community has two groups - the Great Nicobarese and the Little Nicobarese.
- They use different dialects of the Nicobarese language, the Shompen have their own unique language.
- Campbell Bay predominantly houses Great Nicobarese.
- The majority on Great Nicobar comprises people who settled on the island from mainland India. Between 1968 and 1975, the Indian Government settled retired military servicemen and their families from various states.

The population of settlers on the island today is around 6,000.

WHAT ARE THE FEATURES OF NITI AAYOG PROJECT?



- In March 2021, NITI Aayog unveiled a Rs. 72,000 crore plan called “Holistic Development of Great Nicobar Island” at Andaman and Nicobar Islands”.
- It includes the construction of an international transshipment terminal, an international airport, a power plant and a township.
- It is to be implemented by Andaman and Nicobar Islands Integrated Development Corporation (ANIIDCO).

WHAT IS THE NEED FOR TRANSSHIPMENT PORT AND AIRPORT?



- As per Niti Aayog, the proposed transshipment port will allow Great Nicobar to participate in the regional and global maritime economy by becoming a major player in cargo transshipment.
- The proposed airport will support the growth of maritime services and enable Great Nicobar Island to attract international and national visitors to experience the outstanding natural environment and participate in sustainable tourism activity.

The idea is that a transshipment port located near one of the world's busiest international sea routes (the Malacca Strait) will allow increased participation in global maritime trade.

WHAT ARE THE ENVIRONMENTAL CONCERNS?

1 HUGE ECOLOGICAL COSTS

- It has been heavily criticized for its ecological costs and for potential violations of tribal rights.
- The project requires the diversion of about 130 sq. km. of forest land and the felling of around 10 lakh trees.



2 IMPACT ON WILDLIFE

- In January, 2021 the Indian government denotified two wildlife sanctuaries - the Galathea Bay wildlife sanctuary and the Megapode wildlife sanctuary - to make way for the project.
- At the same time, the Government released a National Marine Turtle Action Plan that lists Galathea Bay as a marine turtle habitat in India.
- The transshipment terminal is expected to be developed at Galathea Bay, one of the world's largest nesting sites for the giant leatherback turtle.
- The Giant Leatherback Turtle and the Nicobar Megapode are listed in Schedule I of the Wildlife (Protection Act), 1972

THE IMD ISSUES WARNINGS FOR HEAVY RAINFALL IN A COLOR-CODED SYSTEM.



Heavy Rainfall/ snowfall	No heavy rainfall	Less than 70 mm/hr
	heavy	64.5-115.5 mm/ 24 hrs
	Very heavy	115.6 -204.4 mm/ 24 hrs
	Exceptionally heavy	Greater than or equal to 204.5 mm/ 24 hrs
Thunder storm/squall/ hailstorm	No TS	No Thunderstorm
	Light TS	Thunderstorms with maximum surface wind speed less than 40 kmph (In gusts)
	Moderate TS	Thunderstorms with maximum surface wind speed 41 – 61 kmph (In gusts / squall).
	Severe TS	Thunderstorms with maximum surface wind speed 62 -87 kmph (In gusts/ squall).
	Very Severe TS	Thunderstorms with or without rain with maximum surface wind speed greater than 88 kmph (in gusts/ squall).
	Hailstorm	Severe thunderstorm with hail
Duststorm	No dust storm	No dust storm
	Light dust storm	If the wind speed is up to 40 kmph and visibility is less than 1,000 metres but more than 500 meters
	Moderate dust storm:	If the wind speed is up to 40 kmph and visibility is less than 1,000 metres but more than 500 meters
	Severe dust storm	If surface wind speed (in gusts) is between 62 - 87 kmph (In gusts) and visibility is less than 200 metres
	Very Severe dust storm	If surface wind speed (in gusts) exceeds 88 kmph (In gusts) and visibility is less than 200 metres

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